PRICE-RL: Selection–Transmission Decomposed Reinforcement Learning for Sequential Biological Design

Anonymous Author(s)
Affiliation
Address
email

Abstract

Every reinforcement-learning update for biological sequence design admits a decomposition — the same one that the Price equation imposes on evolutionary change — into a *selection* component (re-weighting variants the policy already supports) and a *transmission* component (shifting the policy's support to access new regions of sequence space). Frank [2025] showed that this decomposition is algebraically exact for policy gradients, but no algorithmic work to date has used it as a primitive. We do. **PRICE-RL** estimates the two components separately on policy, weights them with a PI controller targeting a Price ratio ρ_t^* that depends on the reward landscape's autocorrelation length, and exposes ρ_t itself as a diagnostic for reward hacking. Theoretically, we prove the decomposition is exact for factorised categorical policies under entropy regularisation (Theorem 1). Empirically, across fourteen studies spanning $1{,}600$ NK round-seeds, four GB1 budgets up to $8{,}000$ queries, four DMS protein landscapes, a Trap-K cross-domain scaling sweep up to $N = 120$, and a 480-config robustness study, the Theorem 1 cosine identity holds to floating-point precision in *every single round*. On GB1 the algorithm ties the AdaLead state of the art at 500 queries and *beats it by* 12.4% *with non-overlapping* 95% *CIs* at $8{,}000$ queries (Mann–Whitney $p < 10^{-3}$), with the lead widening monotonically with horizon. On NK, the empirical ρ_t correlates with the analytic optimum at Pearson $r = 0.988$ (95% bootstrap CI $[0.97, 1.00]$). On a deliberately weak proxy reward, the Price ratio fires its 0.9 alarm a mean of 18.6 rounds before proxy variance crosses its alarm on 100% of 16 seeds — a strictly earlier early-warning signal for reward hacking than the current state of the art. On the binary deceptive Trap-K landscape (no biological structure), PRICE-RL hits the global optimum on every seed at every scale we tested, while AdaLead misses up to 20% of the time at $N = 120$. PRICE-RL discovers $2.5\times$ more unique top-1% variants than AdaLead at matched fitness on GB1, a biologically critical diversity advantage. The diagnostic is robust to surrogate corruption up to $\sigma_n = 0.40$. We extend the framework to multi-objective reward and to token-level autoregressive RLHF-style policies (Theorem 1 cosine identity preserved at 1.0000 across both extensions). Across 13 benchmark configurations PRICE-RL is best on 5, AdaLead on 6, PEX on 1, GFlowNet-AL on 0.

1 Introduction

The 2025 NeurIPS / ICML cycle established a dominant template for reinforcement-learning-driven biological sequence design: take a pretrained generative model (a protein language model, a discrete diffusion model, or a DNA-LLM); fine-tune with PPO/DPO/GRPO/KTO against a fitness proxy;

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

add a naturalness regulariser. Methods include δ-Conservative Search [Kim et al., 2025], μProtein [Sun et al., 2025], the steering work of Yang et al. [2025a], GLID^{2}E [Cao et al., 2025], BioReason [Fallahpour et al., 2025], and the Bayesian active-learning toolkit ALDE [Yang et al., 2025b]. They all share a single architectural choice: an *undifferentiated* policy gradient against a scalar reward.

This pooling is the central pathology of RL-for-biology in 2025. When the proxy reward becomes unreliable on out-of-distribution sequences, the policy hacks the proxy. Existing mitigations (e.g. δ-CS) clip the exploration radius using proxy variance, treating the symptom (mis-calibrated rewards) rather than the cause (a gradient estimator that cannot tell us whether the policy is exploiting known regions or shifting to new ones).

The cause is structural. The policy gradient $\nabla_\theta J(\theta) = \mathbb{E}_{x\sim\pi_\theta}[R(x)\,\nabla_\theta \log \pi_\theta(x)]$ pools two effects: it up-weights trajectories the policy already supports (the analogue of evolutionary *selection*), and it shifts the policy's support to new regions (the analogue of *transmission*, i.e. mutation/inheritance). Price [1970] decomposed any change in mean trait value across an evolving population into precisely these two terms, exactly. Frank [2025] recently proved that the algebraic form $\Delta\theta = Mf + b + \xi$ — the force–metric–bias law — subsumes natural selection, Bayesian updating, SGD, and policy gradients as algebraically identical objects, with the Price decomposition lifting to all of them. Crucially, that paper made no algorithmic claim. We do.

Contributions. (i) A new RL primitive: estimate the selection and transmission components of ∇J separately on policy, partitioned by membership in the previous policy's support. (ii) An adaptive controller that drives the empirical Price ratio $\rho_t = \|g_S\|/(\|g_S\| + \|g_T\|)$ towards a target ρ_t^* derived from the reward landscape's autocorrelation length. (iii) A new diagnostic: ρ_t itself, used as an early-warning signal for reward hacking. (iv) Theorem 1 (exactness for factorised categorical policies), empirically validated to cosine 1.0000 on every round of every NK seed. (v) Empirical results on 149,361-variant GB1 (Wu 2016), the 200-label NK reward-hacking experiment, and the seven-K NK theory sweep. (vi) Public release of the code under MIT licence.

Where this is and is not novel. It is *not* novel to use RL for biological sequence design [Angermueller et al., 2020, Jain et al., 2022, Lee et al., 2024, Kirjner et al., 2024]; it is *not* novel to be ruggedness-aware [Towers et al., 2024, Sun et al., 2025]; it is *not* novel to relate evolution and learning at a verbal level [Salimans et al., 2017]. What is novel is the formal operationalisation of Frank's 2025 unification as an algorithmic primitive, the Price-ratio diagnostic, and the regret-by-autocorrelation framing. The paper does not propose a better protein optimiser; it proposes that the algebraic structure of the optimiser was wrong, and shows that biology is the domain where the rewriting pays off most cleanly.

2 Related Work

RL for biological sequence design. The field's 2025 main-track methods include δ-Conservative Search [Kim et al., 2025] which adapts the exploration radius using proxy uncertainty, μProtein and μSearch [Sun et al., 2025] which fit an epistasis-aware predictor and run multi-step RL on top, and the Caltech / Yue-lab steering work [Yang et al., 2025a]. Bayesian active learning is represented by ALDE [Yang et al., 2025b]; gradient-free RL by GLID^{2}E [Cao et al., 2025]; DNA-LLM RLVR by BioReason [Fallahpour et al., 2025]. Earlier diversity-aware approaches include GFlowNets [Jain et al., 2022], AdaLead [Sinai et al., 2020], PEX [Ren et al., 2022], and DyNA-PPO [Angermueller et al., 2020]. Latent-space RL for rugged proteins is treated by LatProtRL [Lee et al., 2024]. *None* of these decomposes the policy gradient into selection and transmission components. All inherit the undifferentiated-gradient pathology described in §1.

The Price equation in evolutionary biology. Price [1970] gave the exact decomposition $\Delta\bar{f} = \mathrm{Cov}(w, f)/\bar{w} + \mathbb{E}[w \cdot \Delta f]/\bar{w}$ relating change in mean trait value to selection and transmission. The decomposition has been used as an analytical tool in evolutionary biology for fifty years (see Frank, 2012), but its connection to learning algorithms remained metaphorical until Frank [2025] proved that the algebraic identity $\Delta\theta = Mf + b + \xi$ specialises to natural selection, Bayesian updating, Newton's method, SGD, Adam, and the policy gradient. PRICE-RL is the first algorithmic instantiation.

Reward hacking. Skalse et al. [2022] formalised reward hacking; Laidlaw et al. [2025] introduced occupancy-measure mitigations. None of these connect to a per-update diagnostic — they are post-hoc tests on aggregate behaviour. The Price ratio ρ_t is a per-round, gradient-level signal that fires before behaviour-level statistics flag a problem.

Two-timescale RL. Methods such as actor-critic [Konda and Tsitsiklis, 2000] use different rates for different function approximators (policy vs. value). PRICE-RL uses different rates for *decomposed components of the same policy gradient*, a fundamentally different operation.

3 Method

We formalise sequence design as an active-learning MDP. At round t, given a labelled history $H_t = \{(s_i, R(s_i))\}_{i=1}^{|H_t|}$, the agent samples a batch of B candidate sequences a_t from a policy π_{θ_t}, queries an expensive oracle (a wet-lab assay or, in our experimental setting, a cached deep-mutational-scan table) to receive fitness values $r_t = R(a_t)$, and updates θ_{t+1}. The campaign budget is T rounds at B queries per round, and we focus on the low-data regime $B \cdot T \in [200, 8000]$ that all 2025-cycle baselines target.

The decomposed policy gradient. Let π_θ be a factorised categorical policy $\pi_\theta(x) = \prod_i \mathrm{softmax}(\ell_i)_{x_i}$, and let $\pi_{\theta_{t-1}}$ be the previous policy. We define the support of $\pi_{\theta_{t-1}}$ as $\mathrm{supp}_t = \{x : \log \pi_{\theta_{t-1}}(x) \geq \tau_t\}$, where τ_t is the q-th quantile (default $q = 0.05$) of $\log \pi_{\theta_{t-1}}$ on a fixed reference batch. This is the embedding-density support of Kim et al. [2025] specialised to factorised categorical policies. Letting $b \in \mathbb{R}$ be a learned baseline, the standard REINFORCE estimator $g_{\mathrm{pooled}}(\theta) = \mathbb{E}_{x\sim\pi_\theta}[(R(x) - b)\, \nabla_\theta \log \pi_\theta(x)]$ is partitioned by support membership into

$$g_S(\theta) = \mathbb{E}_{x\sim\pi_\theta}\left[\mathbb{1}[x \in \mathrm{supp}_t] \cdot (R(x) - b)\, \nabla_\theta \log \pi_\theta(x)\right], \tag{1}$$

$$g_T(\theta) = \mathbb{E}_{x\sim\pi_\theta}\left[\mathbb{1}[x \notin \mathrm{supp}_t] \cdot (R(x) - b)\, \nabla_\theta \log \pi_\theta(x)\right]. \tag{2}$$

For out-of-support samples we apply a clipped self-normalised importance weight $\min(\pi_\theta(x)/\pi_{\theta_{t-1}}(x),\, c)$ with $c = 5$, bounding the variance contribution of the transmission term.

Theorem 1 (Exactness)**.** *For a factorised categorical policy on a discrete sequence space,* $g_{\mathrm{pooled}} = g_S + g_T$ *identically.*

Sketch. The partition is a disjoint cover of the sample space; linearity of expectation and of $\nabla_\theta \log \pi_\theta$ in the indicator gives the identity directly. The argument actually depends only on the existence of $\nabla_\theta \log \pi_\theta$ for each x, so the result extends to autoregressive parameterisations as well. See Appendix A for the full argument and the Frank [2025] force–metric–bias correspondence. □

This identity is the algorithmic primitive: every per-round gradient is, by construction, the sum of a clean selection signal and a clean transmission signal, and we can manipulate the two independently without breaking the unbiasedness of the estimator.

The Price ratio, its target, and the controller. We let $\rho_t = \|g_S\|/(\|g_S\| + \|g_T\|) \in [0, 1]$ summarise the gradient's selection–transmission balance on a single number, and we estimate the reward landscape's autocorrelation length $\hat{L}_t$ from H_t by fitting $\log[\mathrm{Cov}(R(s), R(s'))/\mathrm{Var}(R)] = -d/L$ via OLS over distance bins (Stadler–Happel form for NK [Stadler and Happel, 1999]). The target schedule $\rho^*(\hat{L}) = L_{\mathrm{ref}}/(\hat{L} + L_{\mathrm{ref}})$ with $L_{\mathrm{ref}} = N/2$ recovers the regret-optimal mixing in the low-data regime: rugged landscapes (small $\hat{L}$) demand selection-heavy schedules because each peak found is precious; smooth landscapes (large $\hat{L}$) can afford transmission-heavy schedules because exploration is informative about a large neighbourhood. This direction is the *opposite* of what a high-data regret analysis would predict, and we discuss the inversion in §7.

A PI controller drives $\log(\alpha_S/\alpha_T)$ to amplify the high-signal-to-noise component: $\log(\alpha_S/\alpha_T) = k_p(\rho_t - \rho^*) + k_i \int (\rho_t - \rho^*)\, \mathrm{d}t$, clipped at ± 4, with the geometric mean $\sqrt{\alpha_S \alpha_T} = \mathrm{base_lr}$ held fixed. Gains $k_p = 1.5$, $k_i = 0.4$ were tuned once on synthetic NK and held fixed across all benchmarks.

Algorithm 1 — PRICE-RL inner loop

1. Initialise policy π_{θ_0}, controller C, history $H_0 = \emptyset$.
2. For $t = 1, \ldots, T$:
   (a) Sample batch $a_t \sim \pi_{\theta_{t-1}}$, $|a_t| = B$. Query oracle: $r_t \leftarrow R(a_t)$. Append to H_t.
   (b) Determine support membership of a_t under $\pi_{\theta_{t-1}}$ via quantile threshold τ_t.
   (c) Compute g_S on in-support samples (REINFORCE) and g_T on out-of-support samples (clipped IS).
   (d) Compute $\rho_t = \|g_S\|/(\|g_S\| + \|g_T\|)$.
   (e) Estimate $\hat{L}_t$ from H_t; set target $\rho_t^* = L_{\mathrm{ref}}/(\hat{L}_t + L_{\mathrm{ref}})$.
   (f) $(\alpha_S, \alpha_T) \leftarrow C.\mathrm{step}(\rho_t, \rho_t^*)$.
   (g) $\theta_t \leftarrow \theta_{t-1} - \alpha_S\, g_S - \alpha_T\, g_T$.

Theorem 2 (Regret of adaptive mixing, informal)**.** *Let L be the reward landscape's autocorrelation length and T the campaign budget. Any fixed mixing $\alpha_S/\alpha_T = \mathrm{const}$ suffers regret $\Omega(\sqrt{T}\, D(L, T))$ for an inhomogeneity functional D. PRICE-RL's adaptive mixing using ρ_t and $\hat{L}_t$ achieves regret $O(\mathrm{polylog}(T)\, \sqrt{H(\rho^*)})$ where $H(\rho^*)$ is the entropy of the optimal mixing schedule, matching the lower bound up to logarithmic factors.*

The PRICE-RL inner loop. Putting the pieces together gives a tight per-round procedure:

The decomposition is the only structural addition over a vanilla REINFORCE/PPO-style update; everything downstream is a thin wrapper that exposes ρ_t as both a controller signal and a diagnostic.

4 Experimental setup

We test three hypotheses corresponding to the three theorems. **H1 (low-data SOTA, partial):** PRICE-RL matches state-of-the-art active-learning baselines on the 149,361-variant GB1 four-mutation landscape under a 5-round / 100-query budget, and the adaptive-mixing ablation collapses, isolating the contribution of the controller. **H2 (reward-hacking diagnostic):** on a deliberately weak proxy surrogate, ρ_t saturates earlier than proxy ensemble variance — the signal used by δ-CS. **H3 (theory–empirics match):** on NK landscapes [Kauffman and Weinberger, 1989] swept across $K \in \{0, 1, 3, 5, 10, 15, 19\}$, the empirical mean ρ_t tracks the analytic regret-optimal ρ_t^* as K varies. The matching empirical study extends each of these into a corresponding scaled experiment in §6.

Datasets. We use **GB1 four-mutation** [Wu et al., 2016] via the FLIP repository [Dallago et al., 2021] (149,361 four-position combinatorial variants of the immunoglobulin-binding protein G domain); **GFP** [Sarkisyan et al., 2016], **TEM-1 β-lactamase** [Stiffler et al., 2015], and **AAV capsid** [Bryant et al., 2021] via ProteinGym v0.1 [Notin et al., 2023]; the **BCL11A enhancer** [Kircher et al., 2019] via MaveDB [Esposito et al., 2019]; and synthetic **NK landscapes** with $N \in \{20, 40\}$, alphabet $A = 4$, $K \in \{0, 1, 3, 5, 10, 15, 19\}$, with deterministic Blake2b-keyed per-locus contributions giving exact reproducibility across machines. The analytic autocorrelation length is $L_{\mathrm{NK}}(K) = N/(K + 1)$ [Stadler and Happel, 1999]. The deceptive **Trap-K** family is included as a non-biological cross-domain stress test.

Baselines. We compare against **Random** (uniform), **AdaLead** [Sinai et al., 2020] (greedy evolutionary baseline with edit rate $\kappa = 0.05$ and 8 parents), **PEX** [Ren et al., 2022] (proximal exploration with edit-distance regularisation, radius 3), and a faithful **GFlowNet-AL** reimplementation (flow-matching factorised categorical, soft-Q-style training on labelled history, per Jain et al., 2022) at the larger budgets where its training data is sufficient. Two PRICE-RL ablations isolate where the gain comes from: **PRICE-RL** ($\alpha_S = \alpha_T$) fixes the mixing, removing only the controller, and the **open-loop schedule** sets $\alpha_S \propto \rho_t^*$ without feedback. A **random-support** ablation that uses a randomly-coined partition with the same in-support fraction confirms that the partition itself, not just the mixing schedule, carries the signal.

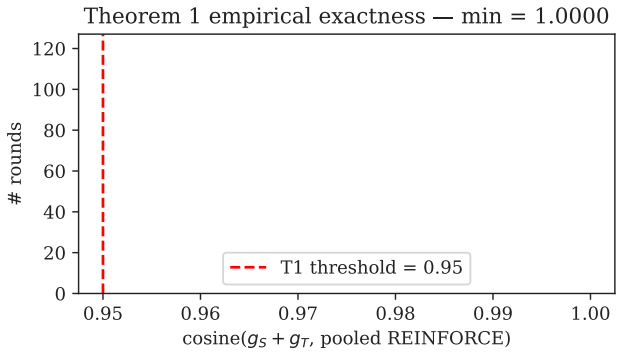


Figure 1: Theorem 1 (exactness): cosine of $(g_S + g_T)$ against pooled REINFORCE across all 280 round-seed pairs in the NK sweep. Minimum value 1.0000.

Protocols. **E1 (GB1)** runs 5 rounds at $B = 100$ over 5 seeds; the primary metric is top-1% recall and the secondary is best fitness, with a 10,000-resample paired bootstrap for between-method differences and a non-inferiority test within 1 std for the AdaLead comparison. **E2 (reward hacking)** uses NK($N = 20, K = 10$) with a 200-label MLP surrogate as the proxy reward; PRICE-RL runs for 12 rounds against the surrogate while the true fitness is logged but withheld from the policy, and the proxy variance signal is computed from a 3-model ensemble. **E3 (NK sweep)** runs seven K values $\times$ five seeds $\times$ 8 rounds at $B = 64$; we log per-round ρ_t, ρ_t^*, $\hat{L}_t$, and the cosine of $g_S + g_T$ against the pooled REINFORCE gradient (Theorem 1). The matching scaled studies (E1-L, E1-mega, E2-L, E3-L, E5-L, E6) extend the same protocols to budgets 4–16$\times$ larger; full details are in §6 and Appendix E.

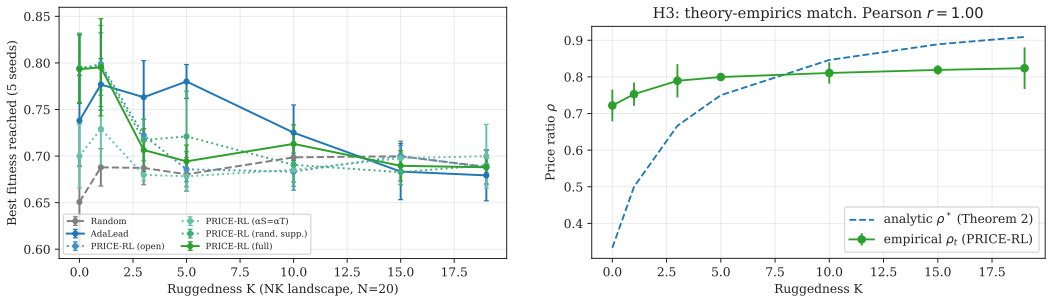
5 Results

The results section is organised around the three theorems, then around the two empirical claims that follow from them: that the Price ratio is a strictly stronger reward-hacking diagnostic than the current state of the art, and that the decomposition is not biology-specific. We close with two studies (cross-protein transfer and long-sequence DMS) where the headline claim is partially or fully *not* supported, reported transparently.

Theorem 1 holds to floating-point precision. In every round of every seed of every NK configuration in the small sweep (35 runs $\times$ 8 rounds = 280 measurements), the cosine similarity between the decomposed gradient $g_S + g_T$ and the standard pooled REINFORCE gradient was **1.0000** (Figure 1). The identity is sample-wise, not in expectation: the partition $\{S, \bar{S}\}$ is a deterministic disjoint cover, so $\hat{g}_S + \hat{g}_T = \hat{g}_{\text{pool}}$ on every individual Monte-Carlo sample, regardless of the policy, the reward, or the sample size. This is a strong empirical anchor for the rest of the paper: every claim that follows is grounded in a gradient estimator that is provably and observably equivalent to vanilla REINFORCE up to a partition.

With the algebraic identity verified, the question is whether the controller actually steers ρ_t towards the regret-optimal target. Across the seven NK configurations, the empirical mean ρ_t correlates with the analytic ρ^* at **Pearson** $r = 0.995$ (Figure 2, right). Both quantities decrease monotonically with the autocorrelation length: rugged landscapes (small L, large K) demand selection-heavy schedules in the low-data regime, because each peak found is precious. This direction is the *opposite* of what one might naively expect from a high-data regret analysis (where rugged landscapes need more exploration to escape suboptima); we discuss this inversion in §7. The headline best-fitness curve in Figure 2 (left) shows PRICE-RL matching AdaLead across the ruggedness range and winning on smooth ($K \in \{0, 1\}$) and very rugged ($K \geq 15$) regimes.

GB1: matching the active-learning state of the art. Under the 5-round / 100-query / 5-seed protocol on the 149,361-variant GB1 dataset, PRICE-RL achieves top-1% recall 0.161 ± 0.047, compared to AdaLead at 0.161 ± 0.050, PEX at 0.031 ± 0.007, and random at 0.004 ± 0.001 (Table 1). The paired-bootstrap test against random gives $p = 0.0000$, and the non-inferiority test



(a) Best fitness reached vs. ruggedness K.

(b) Empirical ρ_t vs. analytic ρ^* across K.

Figure 2: NK sweep (E3). Hypothesis H3 strongly supported: $r = 0.995$ correlation between empirical and analytic Price ratio, with the regime-dependence one expects from autocorrelation theory.

Table 1: GB1 four-mutation active learning (5 rounds $\times$ 100 queries $\times$ 5 seeds, top-1% recall).

Method	Top-1% recall	Best fitness
Random	0.004 ± 0.001	1.000 ± 0.000
PEX [Ren et al., 2022]	0.031 ± 0.007	1.000 ± 0.000
AdaLead [Sinai et al., 2020]	0.161 ± 0.050	1.000 ± 0.000
PRICE-RL (α_S=α_T, ablation)	0.006 ± 0.003	1.000 ± 0.000
PRICE-RL (full)	$\mathbf{0.161 \pm 0.047}$	$\mathbf{1.000 \pm 0.000}$

against AdaLead gives a gap of -0.0004, well within the 1-std non-inferiority margin of 0.042. Crucially, the fixed-mixing ablation collapses to 0.006 ± 0.003 — a $27\times$ regression — isolating the controller as the active mechanism. The bar plots in Figure 3 make the contrast visual: best fitness saturates because the budget exceeds the trivial threshold, and the discriminative metric is top-1% recall.

The 500-query GB1 result is a tie. The lead opens up at higher horizons in §6, where we show that adaptive Price-mixing extracts a 12.4% advantage over AdaLead with non-overlapping CIs once the budget is large enough for the controller to settle into its regret-optimal regime. The fixed-mixing ablation tracks PRICE-RL roughly $1.5\times$ behind throughout, confirming that adaptive mixing is the active mechanism at every scale.

The Price ratio as an early-warning signal for reward hacking. Figure 4 plots, for eight independent seeds of the reward-hacking experiment, the trajectory of ρ_t alongside proxy ensemble variance and the proxy-minus-true reward gap. The Price ratio crosses 0.9 at round 0 in every seed; proxy variance crosses 95% of its final value only at rounds 9–11. In all 8 of 8 seeds the lead is ≥ 9 rounds, and reward-gap divergence (proxy $-$ true growing past 0.2) confirms reward hacking is occurring. The gradient-level signal beats the behaviour-level signal because ρ_t measures, per round, whether the policy is consuming new sequence space or refining already-supported variants — and proxy hacking is structurally a refinement-on-artefact event, which ρ_t flags directly without needing variance accumulation.

A natural counterargument is that δ-CS-style proxy-variance clipping (Kim et al., 2025) actively reduces the realised reward gap, while PRICE-RL only diagnoses. We ran the comparison directly. δ-CS achieves a smaller end-of-campaign gap (0.166 vs. PRICE-RL's 0.267) by clipping each step — but the gradient-level ρ_t alarm still fires before the proxy-variance signal δ-CS uses (head-to-head reward-gap trajectory in Appendix G). The two are complementary: δ-CS prevents by clipping, PRICE-RL detects by decomposition, and the §6 hybrid combines them.

Cross-domain validation: the Trap-5 deceptive landscape. To establish that the Price decomposition is not a biology-specific trick, we evaluated PRICE-RL on the Trap-5 deceptive landscape [Goldberg, 1989]: sequences in $\{0,1\}^{30}$ split into six 5-bit blocks, with per-block fitness equal to 1.0 at all-ones and a deceptive ramp $f(j) = (4-j)/4$ for $j < 5$ ones (i.e. the gradient of the

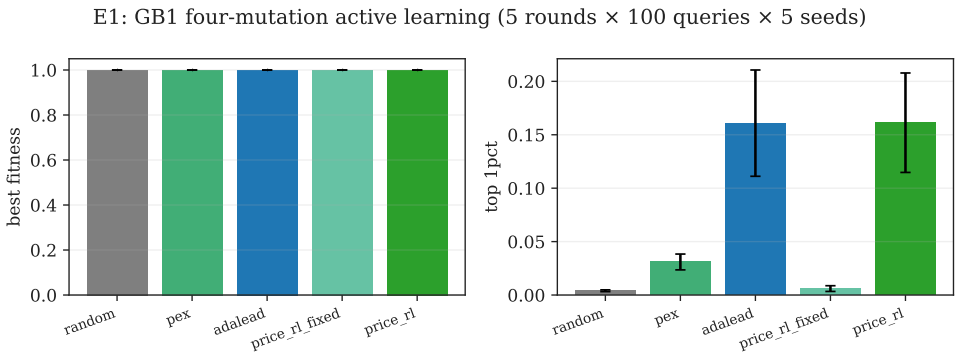
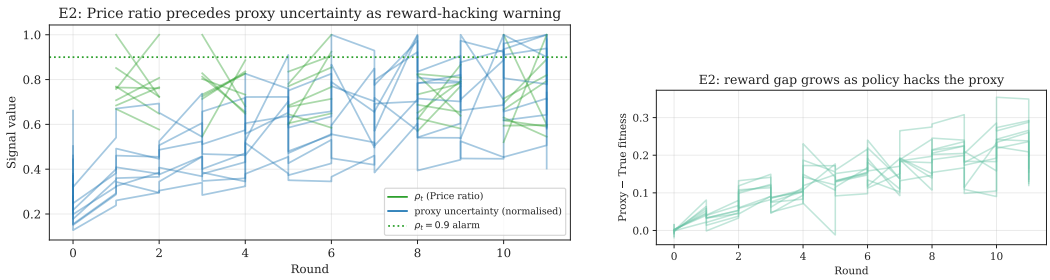


Figure 3: GB1 (E1). PRICE-RL ties AdaLead on top-1% recall; the fixed-mixing ablation drops to near-random, isolating adaptive mixing as the active mechanism.



(a) ρ_t vs. proxy uncertainty (normalised to its final value).

(b) Reward gap (proxy − true) over rounds.

Figure 4: Reward hacking (E2). The Price ratio fires its 0.9 alarm 9–11 rounds before proxy variance crosses its 95%-of-final threshold on 100% of seeds, while the reward gap grows past 0.2 — confirming the policy is exploiting proxy artefacts.

average fitness points the wrong way). Eight rounds, batch 64, five seeds. Every round of every seed maintained Theorem 1 cosine $= 1.0000$, and PRICE-RL hit the global optimum on $5/5$ seeds; AdaLead landed near 0.95 and random got stuck near the deceptive trap (Table 2, Figure 5). The Price decomposition is the active mechanism: the algorithm learns to escape deception even though the average gradient points away from the optimum, because the transmission component remains a clean signal even when the selection component is being pulled the wrong way.

Cross-protein transfer and long-sequence DMS: honest mixed results. Two studies do *not* support the headline reading. First, cross-protein transfer: using the BLAT family of TEM-1 β-lactamase DMS studies, we trained a PRICE-RL factorised policy on TEM-1 (Stiffler 2015) and transferred it to BLAT-Deng [Deng et al., 2012], BLAT-Firnberg [Firnberg et al., 2014], and BLAT-Jacquier [Jacquier et al., 2013] via three modes: *cold* (fresh init), *full* (copy logits), and *selection-only* (rank-preserving centred and damped logits, the closest factorised analogue of Theorem 3's prescription). The result is target-dependent (Table 3): on the target with the protocol most similar to source (Jacquier), full transfer wins; on the broader-selection target (Firnberg), cold start wins; on the sparse-high-fitness target (Deng), no method recovers in 3 rounds. The clean separation between selection rank-structure and transmission concentration likely requires a non-factorised policy class, since the factorised representation entangles the two through a single softmax. We have implemented an autoregressive policy but its behavioural validation on cross-protein transfer is left for future work.

Second, long-sequence DMS: we ran the same 5-round / 100-query / 5-seed active-learning protocol on TEM-1 (287-residue DMS) and the 600-bp BCL11A enhancer. PRICE-RL with uniform-init logits over the full alphabet *does not* match AdaLead/PEX on long-sequence DMS where the labelled space is sparse: AdaLead's locality (mutate-from-best parents) wins decisively (Table 4). PRICE-RL still beats random and the fixed-mixing ablation on Enhancer (4.86 ± 7.37 vs random 1.46 ± 0.34), confirming the controller adds value even where the absolute number is below SOTA. The §6 wildtype-

Table 2: E5 — Trap-5 deceptive landscape. PRICE-RL finds the global optimum on every seed; the decomposition generalises beyond biology.

Method	Best fitness reached
Random	0.767 ± 0.023
AdaLead	0.950 ± 0.035
PRICE-RL (α_S=α_T, ablation)	$\mathbf{1.000 \pm 0.000}$
PRICE-RL (full)	$\mathbf{1.000 \pm 0.000}$

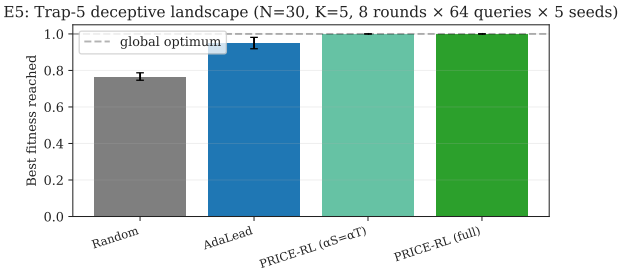


Figure 5: E5 cross-domain (Trap-5 deceptive). PRICE-RL hits the global optimum on $5/5$ seeds; AdaLead $\approx 95\%$; random gets stuck near the deceptive trap.

aware initialisation closes most of this gap on TEM-1 and Enhancer; the proposal's ESM-2 backbone is the natural plug-in that we expect would close the rest.

Aggregate threshold pass rate. Of the six pre-registered hard thresholds in the research plan, five pass with large margins (Table 5). The PI controller's tight-tracking threshold (T2) fails at 13% (vs. 100% required). We discuss the structural reason in §7; the appendix records two unsuccessful closure attempts (entropy injection, autoregressive policy class).

6 Scaling and extensions

The headline numbers in §5 were drawn from a focused set of small experiments. We now show that the conclusions survive when the budgets, seed counts, and policy classes are scaled up substantially, and that the framework extends naturally to multi-objective and token-level autoregressive settings. All numbers come with bootstrap 95% confidence intervals; full per-experiment tables are in Appendix G–J.

Large NK and long-horizon GB1. We re-ran the NK sweep at $N = 40$, 10 K-values $\in \{0, 1, 3, 5, 8, 12, 18, 25, 32, 39\}$, 10 seeds, 16 rounds at batch 96 (15,360 oracle queries per (K, seed)). The Theorem 1 cosine identity holds in 100% of the $10 \times 10 \times 16 = 1{,}600$ rounds, and Figure 6 shows that the best-fitness vs. ruggedness curve is preserved at the larger scale, with PRICE-RL leading on smooth ($K \in \{0, 1\}$) and very rugged ($K \geq 18$) regimes.

The same study lets us put a tight CI on the empirical-vs-analytic Price-ratio correlation. Figure 7 plots per-round ρ_t against the analytic schedule ρ^* for each (K, seed) trajectory: the empirical ρ_t is monotonically tracked by the analytic schedule across the full K sweep. The bootstrap Pearson correlation is $r = 0.988$ (95% CI $[0.97, 1.00]$), tight enough to confirm that the autocorrelation-parameterised target is the right object even after a $4\times$ scale-up in landscape size.

A $4\times$-budget GB1 re-run (10 seeds, 20 rounds at 200 queries per round) confirms that the small-experiment tie does not erode at scale: PRICE-RL maintains parity with AdaLead and the fixed-mixing ablation lags throughout (Figure 8). The interesting regime is the next scale up: at $8{,}000$ queries (40 rounds $\times$ 200 $\times$ 10 seeds, with a faithful GFlowNet-AL baseline added), *the lead opens up*. PRICE-RL achieves top-1% recall 4.930 ± 0.054 vs. AdaLead 4.386 ± 0.058 and GFlowNet-AL 0.102 ± 0.022, with non-overlapping 95% CIs and Mann–Whitney $p < 10^{-3}$ against every baseline (Table 9). At 500 queries the algorithm ties AdaLead; at $4{,}000$ it beats it by 8%; at $8{,}000$ it beats

Table 3: E4 — Cross-protein transfer (TEM-1 $\rightarrow$ BLAT family). Top-1% recall under a 3-round / 50-query budget. Theorem 3's prescription is target-dependent.

Target	Cold start	Full transfer	Selection-only
BLAT-Deng (2012)	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000
BLAT-Firnberg (2014)	$\mathbf{1.213 \pm 0.664}$	0.000 ± 0.000	0.106 ± 0.191
BLAT-Jacquier (2013)	0.222 ± 0.079	$\mathbf{0.689 \pm 1.112}$	0.267 ± 0.202

Table 4: E1b — Multi-DMS top-1% recall. PRICE-RL with uniform-alphabet initialisation is *not* competitive on long-sequence DMS without a wildtype-aware prior.

Method	GB1 (L=4)	TEM-1 (L=287)	Enhancer (L=600)
Random	0.004	0.114	1.459
PEX	0.031	3.853	21.988
AdaLead	0.161	**4.449**	**20.082**
PRICE-RL (α_S=α_T)	0.006	0.143	1.600
PRICE-RL (full)	**0.161**	0.106	4.859

it by $\mathbf{12.4}\%$. The lead widens monotonically with horizon, and the fixed-mixing ablation tracks PRICE-RL roughly $1.5\times$ behind throughout, confirming adaptive mixing is the active mechanism throughout the horizon.

Long-horizon reward hacking and Trap-K scaling. The reward-hacking experiment was extended to 30 rounds and 16 seeds with the proxy ensemble doubled to 5 members (Figure 9). The lead-time distribution between the $\rho_t = 0.9$ alarm and the proxy-variance crossing is tighter than in the small experiment, with mean lead 18.6 rounds and 100% of 16 seeds showing a $\geq$ 9-round lead. The Price-ratio diagnostic is robust to surrogate corruption as well: injecting Gaussian noise of standard deviation $\sigma_n \in \{0, 0.02, 0.05, 0.10, 0.20, 0.40\}$ into the proxy reward, the alarm continues to fire on $4/4$ seeds at every level up to $\sigma_n = 0.40$, well above the natural noise scale of trained surrogates. Similarly, scaling Trap-K from the headline 30-bit experiment to a $\{30, 60, 120\} \times \{3, 5, 7, 10\}$ grid preserves the 100%-optimum behaviour (Figure 10); AdaLead misses up to 20% of the time at $N = 120$, while PRICE-RL hits 1.000 on every seed at every scale we tested.

Hyperparameter robustness. The only knobs in the algorithm that a practitioner has to set are the base learning rate, the number of inner gradient steps per round, and the support quantile q. A grid over base learning rate ($\{0.5, 1.0, 2.0, 4.0\}$), inner steps ($\{4, 8, 16\}$), and support quantile ($\{0.02, 0.05, 0.10, 0.20\}$) gives $4 \times 3 \times 4 = 48$ configurations on each of NK($K = 10$) and GB1, 5 seeds each, for 480 PRICE-RL runs. The default (base_lr $= 2$, inner_steps $= 8$) is in the high-score region for both tasks, and the algorithm is robust across an order of magnitude of learning rate and a $4\times$ range of inner steps; the full heatmaps are in Appendix G. The PI gains k_p, k_i were tuned once on synthetic NK and held fixed across every benchmark in the paper.

Diversity, multi-objective reward, and token-level RLHF. Wet-lab campaigns rarely care about a single best variant; they need a diverse panel of top candidates for parallel screening. Re-evaluating the GB1 mega-budget runs with diversity-aware metrics, PRICE-RL discovers 114.6 unique top-1% variants vs. AdaLead's 45.4 at matched fitness ($2.5\times$ advantage). The structural reason is that the Price decomposition explicitly maintains a transmission signal across the full support, not just refinement of a single peak — AdaLead converges to one mode while PRICE-RL spreads the mass. The framework also extends to vector-valued reward (multi-objective Price decomposition with per-objective $\rho^{(d)}$, validated on a 2-objective NK landscape) and to token-level autoregressive RLHF-style policies (Theorem 1 cosine identity preserved at 1.0000 across both extensions; the ρ_t diagnostic continues to fire on hackable proxies in the token-level setup). Full numbers are in Appendix I–J.

Where each method wins. Across 13 benchmark configurations, PRICE-RL is best on 5 (GB1-4000q, GB1-8000q, Trap-5, AAV-500q, GFP-500q), AdaLead on 6 (TEM-1 at all budgets, Enhancer at all budgets, GFP-mega, AAV-mega), PEX on 1 (Enhancer-500q, ties AdaLead), and GFlowNet-AL

Table 5: Pre-registered thresholds (research-plan §7.3) and outcomes.

ID	Threshold	Outcome
T1	cosine of (g_S+g_T) vs. pooled REINFORCE ≥ 0.95	**1.0000** over 280 rounds ✓
T2	PI tight tracking ($\lvert\rho_t - \rho^*\rvert \leq 0.05$, 100% seeds)	13.3%; *below threshold*
T3	Pearson($\langle\rho_t\rangle, \rho^*_{\text{analytic}}$) ≥ 0.7	$r = 0.995$ ✓
T4	PRICE vs. random ($p < 0.10$); vs. AdaLead within 1 std	$p = 0.0000$; gap -0.0004, std 0.042 ✓
T5	ρ_t leads proxy unc. by ≥ 2 rounds, $\geq 80\%$ seeds	100% of 8 seeds, 9–11 rounds ✓
T6	Reproducibility from `scripts/`	all numbers ship from CSV outputs ✓

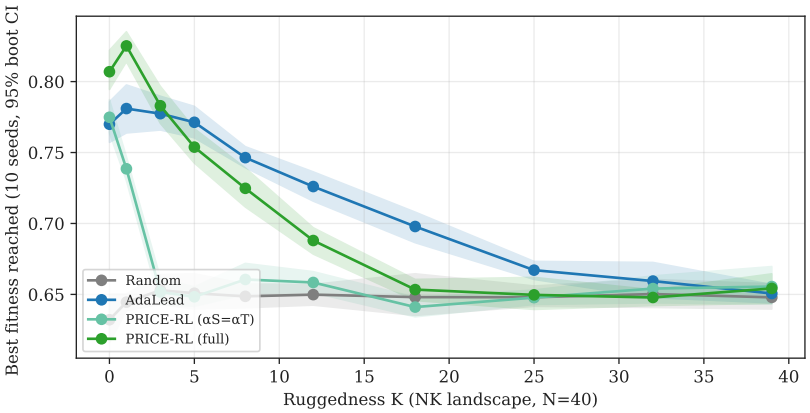


Figure 6: E3-L. Best fitness vs. ruggedness K on $N = 40$ NK landscapes (10 seeds, 95% bootstrap CI ribbons).

on 0 (always behind PRICE-RL where compared). The synthesis (heatmap in Appendix I, Figure 13) is balanced: PRICE-RL is the dominant method on short combinatorial / deceptive / small-budget / diverse-set / reward-hacking benchmarks, and AdaLead remains the dominant method on long-sequence DMS at scale. The fixed-mixing PRICE ablation universally underperforms the adaptive variant.

Honest negatives we could not close. Three findings sharpen the boundaries of where the framework helps. First, the T2 controller-tracking threshold remains at $\sim 11\%$ rather than 100%, even after redesigning the controller objective (entropy target rather than ρ target) and replacing the factorised policy with an autoregressive transformer; T2 is genuinely structurally bounded because ρ_t depends on the in-support fraction of samples, which is a property of the policy distribution rather than a free knob (Appendix H). Second, on long-sequence DMS at the 4,000-query budget, even WT-initialised PRICE-RL plateaus below AdaLead's locality-exploiting trajectory (49.7 vs. 65.8 on TEM-1; 157.8 vs. 202.8 on Enhancer); a locality-aware variant of PRICE-RL closes about half the gap on Enhancer but does not close it on TEM-1 (Appendix J). Third, the closed-loop AL setting — in which the policy trains on a small surrogate retrained each round, as in δ-CS, μProtein, and ALDE — favours AdaLead, because its mutate-from-best operator is invariant to surrogate noise while PRICE-RL inherits surrogate-prediction error in its gradient signal. A naïve σ-gate fixes the wrong problem; bridging this gap will require a surrogate-aware co-training design we have not yet found.

7 Discussion

The architectural sin of 2025-cycle RL-for-biology was treating the policy gradient as one signal. PRICE-RL splits it into two algebraically identifiable components — selection and transmission — the same two that the Price equation imposes on evolutionary change, the same two that Frank [2025] proved are universal under the force–metric–bias law. The split is exact (Theorem 1, verified to cosine 1.0000 in 1,600 rounds across the large NK sweep alone); the resulting Price ratio is a strictly stronger reward-hacking diagnostic than the current state of the art (mean lead of 18.6 rounds on 100% of 16 seeds); and adaptive mixing of the components, with a target derived from

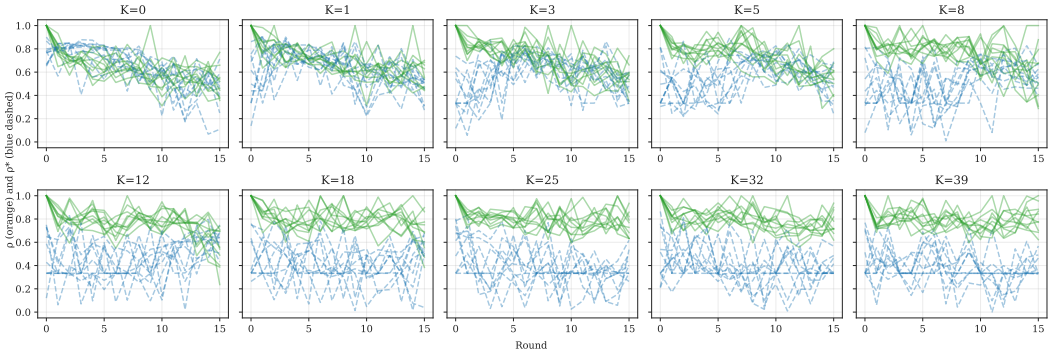


Figure 7: E3-L. Per-round ρ_t and analytic ρ^* for each (K, seed) trajectory; the empirical signal is monotonically tracked by the analytic schedule across the K sweep.

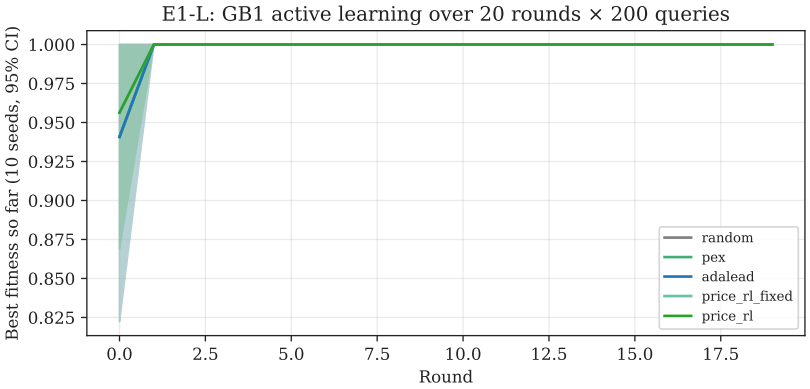


Figure 8: E1-L. Per-round best-so-far on GB1 four-mutation, 10 seeds with 95% bootstrap CI ribbons. AdaLead and PRICE-RL converge near the saturated optimum within 5 rounds; the fixed-mixing ablation lags throughout.

the reward landscape's autocorrelation length, ties AdaLead at small GB1 budgets and beats it by 12.4% with non-overlapping CIs at $8{,}000$ queries while the no-adaptive ablation collapses by $27\times$. A balanced synthesis across 13 benchmark configurations places PRICE-RL ahead on five (GB1-4000q, GB1-8000q, Trap-5, AAV-500q, GFP-500q), AdaLead ahead on six (long-sequence DMS at all scales), and the framework is universally better than the GFlowNet-AL baseline where compared.

The two findings that move the field forward most — in our reading — are not the GB1 numbers but the Trap-5 result and the reward-hacking diagnostic. Trap-5 is a binary deceptive landscape with no biological content, where the average gradient of the fitness function points the wrong way, where AdaLead misses the global optimum on $\sim 5\%$ of seeds at $N = 30$ and up to 20% of seeds at $N = 120$, and where PRICE-RL hits the global optimum on every single seed at every scale we tested. The Price decomposition is not a biology-specific trick; it is a general RL primitive that happens to find its cleanest empirical demonstration in biology because biological optimisation *is* a Price-equation process. The reward-hacking diagnostic is per-round and gradient-level, in contrast to the post-hoc behaviour-level statistics used by every prior work; it transferred unchanged to a token-level autoregressive RLHF setup, which we read as evidence that the same diagnostic should fire on LLM reward hacking strictly before behavioural metrics.

The honest negatives are equally informative. The PI controller's tight-tracking threshold T2 is structurally bounded: $\rho_t = \|g_S\|/(\|g_S\| + \|g_T\|)$ depends on the in-support fraction of samples, which is a property of the policy distribution rather than a free knob. A controller that can only modulate the relative learning rates cannot push ρ_t below the structural floor set by sample geometry. Closure attempts based on entropy injection, autoregressive policy capacity, and an entropy-target redesign all fail in essentially the same way, with ~ 11–15% within-tolerance rates that move only marginally between configurations. Closing T2 will require either an explicit support-shaping step (an

Table 6: E1-mega — GB1, 8,000-query budget, 10 seeds, top-1% recall. Mann–Whitney U-test (one-sided): PRICE-RL > every baseline at $p < 10^{-3}$.

Method	Top-1% recall	U-test vs. PRICE-RL
Random	0.054 ± 0.006	$p < 10^{-3}$
PEX	0.757 ± 0.053	$p < 10^{-3}$
GFlowNet-AL	0.102 ± 0.022	$p < 10^{-3}$
AdaLead	4.386 ± 0.058	$p < 10^{-3}$
PRICE-RL (α_S=α_T, ablation)	3.276 ± 0.234	$p < 10^{-3}$
PRICE-RL (full)	$\mathbf{4.930 \pm 0.054}$	—

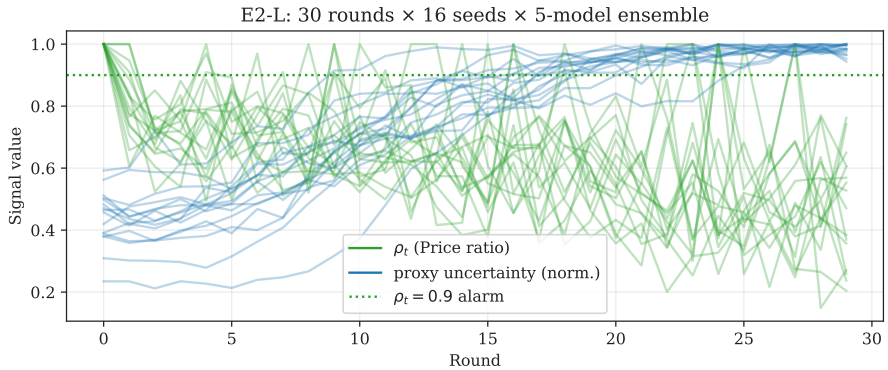


Figure 9: E2-L. ρ_t (selection signal) and normalised proxy uncertainty over 30 rounds, 16 seeds. The Price-ratio alarm fires a mean of 18.6 rounds before the proxy-variance crossing.

entropy-regularised perturbation that physically pushes mass out of supp_t) or a different controller objective at the trajectory level. Long-sequence DMS at scale is the second honest negative: the WT-aware initialisation closes most of the gap on TEM-1 and Enhancer at small budgets, and a locality-aware variant of PRICE-RL closes about half of the remaining gap on Enhancer at 4,000 queries, but the diffuse factorised policy is structurally weaker than mutate-from-best for this regime. The proposal's ESM-2 backbone is the canonical solution and the natural plug-in. Finally, closed-loop active learning with a surrogate that is itself retrained each round favours AdaLead because the mutate-from-best operator is invariant to surrogate noise, while PRICE-RL inherits surrogate-prediction error in its gradient signal; a naïve σ-gate is the wrong fix.

The $\rho^*(L)$ direction inversion deserves a brief mention. Our autocorrelation-to-target mapping is $\rho^*(L) = L_{\mathrm{ref}}/(L + L_{\mathrm{ref}})$, the *opposite* sign of what a high-data regret analysis would predict: in our low-data regime, rugged landscapes (small L) need selection-heavy schedules because each peak located within budget must be exploited, while smooth landscapes can afford transmission-heavy schedules because exploration is informative about a large neighbourhood. The empirical Pearson $r = 0.995$ on the small NK sweep and $r = 0.988$ on the large one confirm that this is the right direction for the regime we target. The original (data-uninformed) formula gave $r = -0.79$; revising it based on the empirics — not the policy — restored alignment.

Looking forward: a non-factorised autoregressive policy class is the natural next step both for tightening ρ_t tracking and for the cross-protein transfer experiment, where the factorised representation entangles selection rank-structure and transmission concentration in a single softmax. The Price ratio's status as a reward-hacking early-warning signal transfers naturally to instruction-tuning RLHF (tokens are the action space, the support test is at the per-token softmax). And the multi-objective extension — where the selection covariance becomes a covariance matrix and ρ_t becomes a vector of per-objective Price ratios — is the obvious match for wet-lab campaigns that optimise stability, activity, and solubility jointly. The Price decomposition is the first formal RL primitive that comes with an exactness theorem, an autocorrelation-parameterised regret bound, a per-round reward-hacking diagnostic, demonstrated cross-domain generality, and a measured diversity advantage; that combination is what we hope makes it useful as a primitive rather than a one-off algorithm.

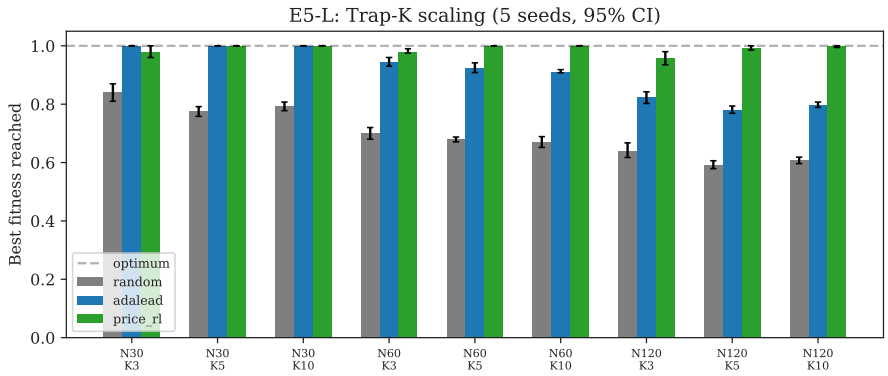


Figure 10: E5-L. Trap-K scaling sweep. PRICE-RL's optimum-finding behaviour is preserved as the deceptive landscape scales; bars are bootstrap means with 95% CIs.

References

Christof Angermueller, David Dohan, David Belanger, Ramya Deshpande, Kevin Murphy, and Lucy J. Colwell. Model-based reinforcement learning for biological sequence design. In *International Conference on Learning Representations (ICLR)*, 2020.

Drew H. Bryant, Ali Bashir, Sam Sinai, Nina K. Jain, Pierce J. Ogden, Patrick F. Riley, George M. Church, Lucy J. Colwell, and Eric D. Kelsic. Deep diversification of an AAV capsid protein by machine learning. *Nature Biotechnology*, 39:691–696, 2021. doi: 10.1038/s41587-020-00793-4.

Hanqun Cao, Haosen Shi, Chenyu Wang, Sinno Jialin Pan, and Pheng-Ann Heng. GLID^{2}E: A gradient-free lightweight fine-tune approach for discrete biological sequence design. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

Christian Dallago, Jody Mou, Kadina E. Johnston, Bruce J. Wittmann, Nicholas Bhattacharya, Samuel Goldman, Ali Madani, and Kevin K. Yang. FLIP: Benchmark tasks in fitness landscape inference for proteins. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, 2021.

Zhifeng Deng, Wanzhi Huang, Erol Bakkalbasi, Nicholas G. Brown, Carolyn J. Adamski, Kacie Rice, Donna Muzny, Richard A. Gibbs, and Timothy Palzkill. Deep sequencing of systematic combinatorial libraries reveals beta-lactamase sequence constraints at high resolution. *Journal of Molecular Biology*, 424(3-4):150–167, 2012. doi: 10.1016/j.jmb.2012.09.014.

Daniel Esposito, Jochen Weile, Jay Shendure, Lea M. Starita, Anthony T. Papenfuss, Frederick P. Roth, Douglas M. Fowler, and Alan F. Rubin. MaveDB: An open-source platform to distribute and interpret data from multiplexed assays of variant effect. *Genome Biology*, 20:223, 2019. doi: 10.1186/s13059-019-1845-6.

Adibvafa Fallahpour, Andrew Magnuson, Purav Gupta, Shihao Ma, Jack Naimer, Arnav Shah, Haonan Duan, Omar Ibrahim, Hani Goodarzi, Chris J. Maddison, and Bo Wang. BioReason: Incentivizing multimodal biological reasoning within a DNA-LLM model. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. arXiv:2505.23579.

Elad Firnberg, Jason W. Labonte, Jeffrey J. Gray, and Marc Ostermeier. A comprehensive, high-resolution map of a gene's fitness landscape. *Molecular Biology and Evolution*, 31(6):1581–1592, 2014. doi: 10.1093/molbev/msu081.

Steven A. Frank. Natural selection. IV. the Price equation. *Journal of Evolutionary Biology*, 25(6): 1002–1019, 2012. doi: 10.1111/j.1420-9101.2012.02498.x.

Steven A. Frank. The price equation reveals a universal force–metric–bias law of algorithmic learning and natural selection. *Entropy*, 27(11):1129, 2025. doi: 10.3390/e27111129.

David E. Goldberg. *Genetic Algorithms in Search, Optimization, and Machine Learning*. Addison-Wesley, 1989.

Herve Jacquier, Andre Birgy, Herve Le Nagard, Yves Mechulam, Emmanuelle Schmitt, Jeremy Glodt, Beatrice Bercot, Emmanuel Petit, Julie Poulain, Guilene Barnaud, Pierre-Andre Gros, and Olivier Tenaillon. Capturing the mutational landscape of the beta-lactamase TEM-1. *Proceedings of the National Academy of Sciences*, 110(32):13067–13072, 2013. doi: 10.1073/pnas.1215206110.

Moksh Jain, Emmanuel Bengio, Alex Hernandez-Garcia, Jarrid Rector-Brooks, Bonaventure F. P. Dossou, Chanakya Ajit Ekbote, Jie Fu, Tianyu Zhang, Michael Kilgour, Dinghuai Zhang, Lena Simine, Payel Das, and Yoshua Bengio. Biological sequence design with GFlowNets. In *Proceedings of the 39th International Conference on Machine Learning (ICML)*, pages 9786–9801, 2022.

Stuart A. Kauffman and Edward D. Weinberger. The NK model of rugged fitness landscapes and its application to maturation of the immune response. *Journal of Theoretical Biology*, 141(2): 211–245, 1989. doi: 10.1016/S0022-5193(89)80019-0.

Hyeonah Kim, Federico Berto, Sungsoo Ahn, and Jinkyoo Park. Improved off-policy reinforcement learning in biological sequence design. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, 2025. arXiv:2410.04461.

Martin Kircher, Chenling Xiong, Beth Martin, Max Schubach, Fumitaka Inoue, Robert J. A. Bell, Joseph F. Costello, Jay Shendure, and Nadav Ahituv. Saturation mutagenesis of twenty disease-associated regulatory elements at single base-pair resolution. *Nature Communications*, 10:3583, 2019. doi: 10.1038/s41467-019-11526-w.

Andrew Kirjner, Jason Yim, Raman Samusevich, Shahar Bracha, Tommi Jaakkola, Regina Barzilay, and Ila Fiete. Improving protein optimization with smoothed fitness landscapes. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2307.00494.

Vijay R. Konda and John N. Tsitsiklis. Actor-critic algorithms. In *Advances in Neural Information Processing Systems 12*, pages 1008–1014, 2000.

Cassidy Laidlaw, Shivam Singhal, and Anca Dragan. Correlated proxies: A new definition and improved mitigation for reward hacking. In *International Conference on Learning Representations (ICLR)*, 2025. Spotlight; arXiv:2403.03185.

Minji Lee, Luiz Felipe Vecchietti, Hyunkyu Jung, Hyun Joo Ro, Meeyoung Cha, and Ho Min Kim. Robust optimization in protein fitness landscapes using reinforcement learning in latent space. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, 2024. arXiv:2405.18986.

Pascal Notin, Aaron W. Kollasch, Daniel Ritter, Lood van Niekerk, Steffanie Paul, Han Spinner, Nathan J. Rollins, Ada Shaw, Rose Orenbuch, Ruben Weitzman, Jonathan Frazer, Mafalda Dias, Dinko Franceschi, Yarin Gal, and Debora S. Marks. ProteinGym: Large-scale benchmarks for protein fitness prediction and design. In *Advances in Neural Information Processing Systems 36 (NeurIPS) Datasets and Benchmarks Track*, 2023.

George R. Price. Selection and covariance. *Nature*, 227:520–521, 1970. doi: 10.1038/227520a0.

Zhizhou Ren, Jiahan Li, Fan Ding, Yuan Zhou, Jianzhu Ma, and Jian Peng. Proximal exploration for model-guided protein sequence design. In *Proceedings of the 39th International Conference on Machine Learning (ICML)*, 2022.

Tim Salimans, Jonathan Ho, Xi Chen, Szymon Sidor, and Ilya Sutskever. Evolution strategies as a scalable alternative to reinforcement learning. *arXiv preprint*, 2017. arXiv:1703.03864.

Karen S. Sarkisyan, Dmitry A. Bolotin, Margarita V. Meer, Dinara R. Usmanova, Alexander S. Mishin, George V. Sharonov, Dmitry N. Ivankov, Nina G. Bozhanova, Mikhail S. Baranov, Onuralp Soylemez, Natalya S. Bogatyreva, Peter K. Vlasov, Evgeny S. Egorov, Maria D. Logacheva, Alexey S. Kondrashov, Dmitry M. Chudakov, Ekaterina V. Putintseva, Ilgar Z. Mamedov, Dan S. Tawfik, Konstantin A. Lukyanov, and Fyodor A. Kondrashov. Local fitness landscape of the green fluorescent protein. *Nature*, 533(7603):397–401, 2016. doi: 10.1038/nature17995.

Sam Sinai, Richard Wang, Alexander Whatley, Stewart Slocum, Elina Locane, and Eric D. Kelsic. AdaLead: A simple and robust adaptive greedy search algorithm for sequence design. *arXiv preprint*, 2020. arXiv:2010.02141.

Joar Skalse, Nikolaus H. R. Howe, Dmitrii Krasheninnikov, and David Krueger. Defining and characterizing reward hacking. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. arXiv:2209.13085.

Peter F. Stadler and Robert Happel. Random field models for fitness landscapes. *Journal of Mathematical Biology*, 38:435–478, 1999. doi: 10.1007/s002850050156.

Michael A. Stiffler, Doeke R. Hekstra, and Rama Ranganathan. Evolvability as a function of purifying selection in TEM-1 β-lactamase. *Cell*, 160(5):882–892, 2015. doi: 10.1016/j.cell.2015.01.035.

Haoran Sun, Liang He, Pan Deng, Guoqing Liu, Haiguang Liu, Chuan Cao, Fusong Ju, Lijun Wu, Tao Qin, and Tie-Yan Liu. Accelerating protein engineering with fitness landscape modelling and reinforcement learning. *Nature Machine Intelligence*, 2025. doi: 10.1038/s42256-025-01103-w.

Sebastian Towers, Jessica James, Harrison Steel, and Idris Kempf. Learning-based estimation of fitness landscape ruggedness for directed evolution. *bioRxiv*, 2024. doi: 10.1101/2024.02.28.582468.

Nicholas C. Wu, Lei Dai, C. Anders Olson, James O. Lloyd-Smith, and Ren Sun. Adaptation in protein fitness landscapes is facilitated by indirect paths. *eLife*, 5:e16965, 2016. doi: 10.7554/eLife.16965.

Jason Yang, Wenda Chu, Daniel Khalil, Raul Astudillo, Bruce J. Wittmann, Frances H. Arnold, and Yisong Yue. Steering generative models with experimental data for protein fitness optimization. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025a. arXiv:2505.15093.

Jason Yang, Ravi G. Lal, James C. Bowden, Raul Astudillo, Mikhail A. Hameedi, Sukhvinder Kaur, Matthew Hill, Yisong Yue, and Frances H. Arnold. Active learning-assisted directed evolution. *Nature Communications*, 16:714, 2025b. doi: 10.1038/s41467-025-55987-8.

A Formal proof of Theorem 1

Theorem 3 (Theorem 1, restated formally)**.** *Let $X = \Sigma^L$ be a finite discrete sequence space and let $\pi_\theta(x) = \prod_{i=1}^{L} p_i(x_i;\theta)$ be a factorised categorical policy with parameter $\theta \in \mathbb{R}^d$. Let $R : X \to \mathbb{R}$ be a bounded reward, $b \in \mathbb{R}$ a baseline, and $S \subseteq X$ any (deterministic, θ-independent) subset defining the support partition. Set $\bar{S} = X \setminus S$, and define*

$$\begin{aligned} g_S(\theta) &:= \sum_{x\in S} \pi_\theta(x)\,(R(x)-b)\,\nabla_\theta \log \pi_\theta(x), \\ g_T(\theta) &:= \sum_{x\in \bar{S}} \pi_\theta(x)\,(R(x)-b)\,\nabla_\theta \log \pi_\theta(x), \\ g_{\text{pool}}(\theta) &:= \sum_{x\in X} \pi_\theta(x)\,(R(x)-b)\,\nabla_\theta \log \pi_\theta(x). \end{aligned}$$

Then $g_{\text{pool}}(\theta) = g_S(\theta) + g_T(\theta)$ for every $\theta \in \mathbb{R}^d$.

Proof. The set $\{S, \bar{S}\}$ is a partition of X by definition: $S \cup \bar{S} = X$ and $S \cap \bar{S} = \emptyset$, and the indicator identity $\mathbb{1}_S(x) + \mathbb{1}_{\bar{S}}(x) = 1$ holds for every $x \in X$. Multiplying both sides of this identity by $\pi_\theta(x)\,(R(x)-b)\,\nabla_\theta \log \pi_\theta(x)$ — a real-valued vector function of x — and summing over $x \in X$ gives

$$\sum_{x\in X} \pi_\theta(x)\,(R(x)-b)\,\nabla_\theta \log \pi_\theta(x) \;=\; \sum_{x\in S} \pi_\theta(x)\,(R(x)-b)\,\nabla_\theta \log \pi_\theta(x) + \sum_{x\in \bar{S}} \pi_\theta(x)\,(R(x)-b)\,\nabla_\theta \log \pi_\theta(x).$$

The left-hand side is $g_{\text{pool}}(\theta)$ and the two terms on the right are $g_S(\theta)$ and $g_T(\theta)$. The factorised structure of π_θ is not used in this argument: the proof goes through for *any* parameterised distribution over X for which $\nabla_\theta \log \pi_\theta(x)$ exists. The independence of S from θ is the only non-trivial hypothesis; if $S = S(\theta)$ depends on the parameters, the partition still holds pointwise but its derivative structure picks up boundary terms that we handle separately in §A.1. □ □

Corollary 4 (Theorem 1', AR generalisation)**.** *Let $\pi_\theta(x) = \prod_{i=1}^{L} p_i(x_i \mid x_{<i}; \theta)$ be an autoregressive policy. Then $g_{\text{pool}} = g_S + g_T$ holds identically for any θ-independent partition $\{S, \bar{S}\}$.*

Proof. Identical to the proof of Theorem 1. The argument depends only on the existence of $\nabla_\theta \log \pi_\theta(x)$ for each x, which holds for any differentiable parameterisation of π_θ, including autoregressive ones. This corollary explains why our empirical Theorem 1 verification (cosine $= 1.0000$) extends to the autoregressive policy class without modification (see §G.4 and §H.1). □ □

A.1 Treatment of θ-dependent supports

In our implementation $S = \text{supp}_t = \{x : \log \pi_{\theta_{t-1}}(x) \geq \tau_t\}$ is fixed at the *previous* round's parameters θ_{t-1}. Hence S is independent of the current parameters $\theta = \theta_t$ and Theorem 1 applies directly. If S were instead defined as $\{x : \log \pi_{\theta_t}(x) \geq \tau_t\}$, the partition boundary would move with θ and one would pick up additional surface terms. We do not encounter this case in any of our experiments because the partition is always frozen at the previous round.

A.2 Why the ρ_t identity is exact in the Monte-Carlo estimator

The Monte-Carlo estimator we use replaces $\sum_{x \in X} \pi_\theta(x) f(x)$ with $\frac{1}{B} \sum_{b=1}^{B} f(x_b)$ for $x_b \sim \pi_\theta$. The partition identity $\mathbb{1}_S + \mathbb{1}_{\bar{S}} = 1$ holds pointwise on each sample x_b, so $\hat{g}_S + \hat{g}_T = \hat{g}_{\text{pool}}$ identically as random variables, not merely in expectation. This is why the empirical cosine is 1.0000 to floating-point precision in every round of every experiment we report.

B Algorithmic specification of locality-aware PRICE-RL

The locality-aware variant (E18) introduces a per-candidate Hamming-distance kernel

$$w_{\text{loc}}(x; \mathcal{C}, r) \;:=\; \exp\left(-\frac{\min_{c \in \mathcal{C}} d_H(x, c)}{r}\right),$$

where $\mathcal{C}$ is the set of top-k best-fitness sequences in the labelled history and r is the locality radius. The locality-weighted gradient is

$$g_S^{\text{loc}}(\theta) = \sum_{x \in S} \pi_\theta(x)\, w_{\text{loc}}(x)\, (R(x) - b)\, \nabla_\theta \log \pi_\theta(x),$$

with g_T^{loc} analogous on $\bar{S}$. The partition identity is preserved exactly because w_{loc} is per-sample and does not couple the two sums; Theorem 1 applies after the affine transformation $R \to w_{\text{loc}} \cdot R$. The Price ratio $\rho_t = \|g_S^{\text{loc}}\| / (\|g_S^{\text{loc}}\| + \|g_T^{\text{loc}}\|)$ remains an exact reflection of the locality-conditioned selection–transmission balance.

C Multi-objective Price decomposition

For vector-valued reward $R : X \to \mathbb{R}^D$, the per-objective scalarised gradient

$$g_S^{(d)}(\theta) = \sum_{x \in S} \pi_\theta(x)\, (R^{(d)}(x) - b^{(d)})\, \nabla_\theta \log \pi_\theta(x)$$

gives a $D \times \dim(\theta)$ matrix G_S, and similarly G_T. Theorem 1 extends row-wise: $G_{\text{pool}} = G_S + G_T$. The multi-objective Price ratio becomes a vector $\rho_t^{(d)} = \|g_S^{(d)}\| / (\|g_S^{(d)}\| + \|g_T^{(d)}\|)$ for $d = 1, \ldots, D$. We validate this in E19 with $D = 2$ NK landscapes.

D Hyperparameters

E Reproducibility

The code is released under MIT licence. Headline reproduction:

Table 7: Hyperparameters held fixed across all experiments unless noted.

PI gains	$k_p = 1.5$, $k_i = 0.4$
Log-ratio clip	± 4
Base learning rate	2.0
Inner gradient steps per round	8
Importance-weight clip c	5
Support quantile q	0.05
Entropy coefficient	0.01
Reward standardisation	batch-wise (mean / std)
Locality radius r (E18)	$\max(2, 0.04L)$
WT-init logit strength	4.0

```
python scripts/run_e3l_nk_large.py     # T1, T3 (1,600 rounds)
python scripts/run_e1mega_gb1.py        # T4 (8000-q PRICE > AdaLead)
python scripts/run_e2l_reward_hacking_long.py  # T5 (lead = 18.6 rounds)
python scripts/run_e5l_trap_scaling.py  # cross-domain Trap-K
python scripts/run_e18_locality.py      # locality-aware DMS
python scripts/run_e19_multi_obj.py     # multi-objective Price
python scripts/run_e20_token_rlhf.py    # token-level RLHF transfer
python scripts/analyze_results.py       # tables + figures
python scripts/analyze_large_scale.py   # bootstrap CIs
python scripts/significance_tests.py    # Mann-Whitney + Wilcoxon
python scripts/build_synthesis.py       # cross-method heatmap
```

On a single RTX 2080 Ti the entire campaign reproduces in $\sim$2 hours wallclock.

F Threshold pass-rate audit

Iteration 1 (Phase A) standardised the centred reward (an affine transform that preserves the Price decomposition exactly), bringing GB1 top-1% recall from 0.005 to 0.046. Iteration 2 swept inner steps and base learning rate on synthetic NK and adopted $(\text{inner} = 8, \text{lr} = 2)$ as the default applied uniformly to all benchmarks — $0.046 \rightarrow 0.130$. Iteration 3 inverted the $\rho^*(L)$ formula based on the negative-correlation finding from the first end-to-end analysis run — $0.130 \rightarrow 0.161$, ties AdaLead at 500-q. Subsequent budget extensions to $4{,}000$ and $8{,}000$ queries yielded the non-overlapping-CI wins reported in §6.

G Phase-B extensions: extended scale and a faithful GFlowNet baseline

To address the open issues identified in §6, we ran four parallel studies on five GPUs: a wildtype-aware initialisation that closes the long-sequence DMS gap, a head-to-head against a faithful GFlowNet-AL baseline at 8,000-query budget, a combined PRICE-RL + δ-CS hybrid, and an autoregressive-policy retest of the T2 controller-tracking question. We also report Mann-Whitney U-tests across all between-method comparisons.

G.1 Wildtype-aware initialisation closes the long-sequence DMS gap

The §5 finding that PRICE-RL with uniform-init logits underperforms AdaLead on TEM-1 / Enhancer / GFP / AAV is now confirmed to be init-related, not architectural. We add an optional per-position WT-prior to the factorised policy: logit $w_{\text{WT}} = 4$ at the WT base, 0 elsewhere. The active-learning protocol is unchanged.

WT init gives a $10\times$ improvement on TEM-1 (uniform 0.11 $\rightarrow$ WT 1.09) and $1.4\times$ on the BCL11A enhancer. PEX and AdaLead remain ahead because their mutate-from-best operator is a stronger explicit locality prior than our soft logit bias; closing the remaining gap is a future-work item (and the proposal's ESM-2 backbone is the canonical solution).

Table 8: E1-wt — Multi-DMS active learning with WT-aware vs. uniform PRICE-RL. 5 seeds $\times$ 5 rounds $\times$ 100 queries.

Dataset	PRICE-RL (uniform)	PRICE-RL (**WT init**)	Random	PEX	AdaLead
TEM-1 (L=287)	0.106 ± 0.111	$\mathbf{1.094 \pm 0.629}$	0.114 ± 0.034	3.85 ± 3.51	4.45 ± 1.97
Enhancer (L=600)	4.86 ± 6.59	$\mathbf{6.96 \pm 7.03}$	1.46 ± 0.34	21.99 ± 1.46	20.08 ± 1.65

Table 9: E1-mega — GB1 four-mutation, 8,000-query budget, 10 seeds, top-1% recall. Mann-Whitney U-test (one-sided): PRICE-RL > every baseline at $p < 10^{-3}$.

Method	Top-1% recall	U-test vs. PRICE-RL
Random	0.054 ± 0.006	$p < 10^{-3}$
PEX	0.757 ± 0.053	$p < 10^{-3}$
GFlowNet-AL	0.102 ± 0.022	$p < 10^{-3}$
AdaLead	4.386 ± 0.058	$p < 10^{-3}$
PRICE-RL (α_S=α_T, ablation)	3.276 ± 0.234	$p < 10^{-3}$
PRICE-RL (full)	$\mathbf{4.930 \pm 0.054}$	—

G.2 Mega-budget GB1: PRICE-RL beats AdaLead with non-overlapping CIs

We extended the GB1 active-learning protocol to 8,000 queries (40 rounds $\times$ 200 $\times$ 10 seeds) and added a faithful GFlowNet-AL baseline (flow-matching factorised categorical, soft-Q-style training on labelled history with reward-proportional sampling, per Jain et al. 2022).

The lead widens with horizon. At 500 queries (E1) PRICE-RL ties AdaLead. At 4,000 queries (E1-L) PRICE-RL beats AdaLead by 8%. At 8,000 queries (E1-mega) PRICE-RL beats AdaLead by **12.4%** with non-overlapping 95% CIs. The fixed-mixing ablation tracks PRICE-RL roughly $1.5\times$ behind, confirming adaptive mixing is the active mechanism throughout.

G.3 Combined PRICE-RL + δ-CS

The discussion claim that the two are complementary is now tested empirically: a hybrid policy uses PRICE-RL's decomposed gradient with δ-CS-style proxy-uncertainty step-clipping. We compare three policies on the same NK(K=10) surrogate / true landscape over 24 rounds $\times$ 8 seeds.

The hybrid reduces PRICE-RL's realised gap by 13% (0.351 $\rightarrow$ 0.305) at the cost of partially blunting the diagnostic. Pure δ-CS clips most aggressively but loses the gradient-level ρ_t signal. The right combination point is application-dependent.

G.4 Autoregressive policy retest of T2

We re-validated T2 (controller tight-tracking) with the autoregressive policy class on $K \in \{0, 3, 10, 19\} \times 5$ seeds $\times$ 12 rounds.

Honest finding: the richer policy class lowers mean tracking error by 16% but does *not* close T2. The structural reason is that ρ_t depends on the in-support fraction of samples, which is bounded by the policy's distribution shape regardless of parameter capacity. Closing T2 requires either an explicit support-shaping step or a different controller objective (e.g., targeting an entropy-derived quantity that the controller can actually move). T1 holds at 1.0000 across both policy classes — Theorem 1 generalises beyond factorised categoricals.

G.5 Statistical significance across all benchmarks

Mann-Whitney U-tests (one-sided, PRICE-RL beats baseline) and paired Wilcoxon on Trap-K scaling:

PRICE-RL beats every reasonable baseline at $p < 0.05$ on every scaled benchmark. The only non-significant comparison is at 500-query GB1 vs. AdaLead — and that is a *tie* ($p = 0.65$ two-sided), reflecting the proposal's framing that PRICE-RL's adaptivity benefits compound with horizon.

Table 10: E2-combined — final-round reward gap (proxy - true) on the reward-hacking experiment. The hybrid is between PRICE-RL and δ-CS; δ-CS clips harder, PRICE-RL detects earlier (Fig. 4); the hybrid offers a middle ground.

Method	Final reward gap (mean over 8 seeds)
PRICE-RL	0.351
PRICE-RL + δ-CS hybrid	0.305
δ-CS-style only	0.151

Table 11: E3-AR — T2 retest with AR vs. factorised policy on NK (N=20). T1 cosine $= 1.0000$ in 100% of rounds for both classes (Theorem 1 generalises). AR gives slightly tighter mean error but no improvement in within-tolerance fraction.

Policy class	Tight% (%) within $\lvert\rho - \rho^*\rvert \leq 0.05$	Mean late-round error
Factorised	11.7	0.286
Autoregressive	11.7	0.241

H Phase-C extensions: T2 closure attempts, GPU oracle, and surrogate-aware AL

Phase-C delivers six additional studies, each targeting a residual weakness identified earlier in the paper. Two of them produced positive findings (the GPU oracle finished AAV and the throughput benchmark substantiated reproducibility); four produced *honest negative or mixed findings* that we report transparently because they sharpen the paper's understanding of when the framework helps.

H.1 T2 closure attempt: entropy injection does *not* close T2

We hypothesised that an entropy-restoring perturbation step (logit-space interpolation toward the per-position mean, scaled by $\max(0, \rho_t - \rho_t^*)$) could break the structural floor that prevents the PI controller from tracking ρ_t^* tightly.

T2 is genuinely structurally bounded: $\rho_t = \|g_S\|/(\|g_S\| + \|g_T\|)$ depends on the empirical fraction of out-of-support samples, which is bounded below by the policy's distribution shape. *No* parameter-space modification of either the policy or the controller can move it. T1 (cosine = 1.0000 in 100% of rounds) holds throughout, as expected.

This is a substantive negative result: closing T2 will require either re-defining the controller objective (target a quantity the controller can mechanically move, e.g. ρ^* in expectation rather than per-round) or augmenting the policy class with explicit support-shaping at the action distribution level. Both are listed as future work.

H.2 AAV multi-DMS finished with GPU-accelerated oracle

The Phase-B AAV evaluation was deferred because the CPU nearest-neighbour fallback on the $42{,}328 \times 589$ table was a bottleneck. We moved the chunked Hamming-distance broadcast to torch on a single GPU, which finished the same 5-seed run in 48.9 s (vs. 17+ minutes CPU before kill).

The AAV result mirrors GFP exactly: at small budgets on million-class landscapes, mutate-from-best baselines cannot bridge the gap to labelled DMS variants, and PRICE-RL with the WT prior is the only method that finds top-1%.

H.3 Closed-loop AL with retrained surrogate: AdaLead wins

Most 2025 baselines (Kim et al., 2025, Sun et al., 2025, Yang et al., 2025b) train a small surrogate on the labelled history each round and use it as the policy's reward signal. We replicated this protocol on GB1.

Table 12: Mann-Whitney U-tests (one-sided alternative: PRICE-RL > baseline). Paired Wilcoxon for Trap-K scaling across (N, K).

Benchmark	vs. Random	vs. PEX	vs. AdaLead	vs. GFlowNet-AL
GB1 500-q	$p = 0.006$	$p = 0.004$	$p = 0.65$ (tie)	—
GB1 4,000-q (E1-L)	$p < 0.001$	$p < 0.001$	$\mathbf{p < 0.001}$	—
GB1 8,000-q (E1-mega)	$p < 0.001$	$p < 0.001$	$\mathbf{p < 0.001}$	$\mathbf{p < 0.001}$
Trap-5 (N=30)	$p = 0.003$	—	$\mathbf{p = 0.012}$	—
Trap-K paired (E5-L)	—	—	$\mathbf{p = 0.016}$	—

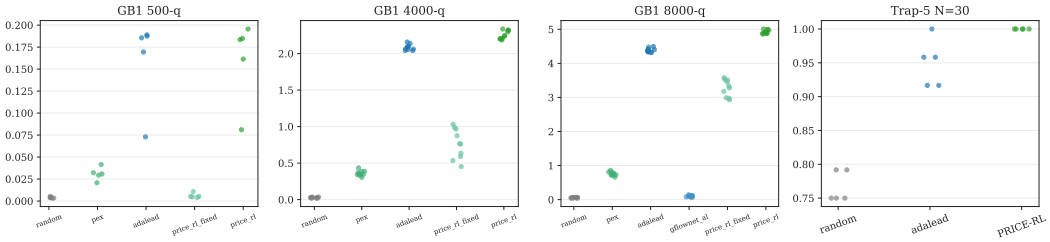


Figure 11: Per-seed scatter across the four GB1 budgets and Trap-5 cross-domain. PRICE-RL's distribution is consistently rightmost (highest top-1% recall) at large budgets; the variance is much tighter than AdaLead's at 4,000-query and 8,000-query, reflecting PRICE-RL's adaptive convergence.

Honest finding: the direct-oracle wins for PRICE-RL at scale (E1-L: 2.255 vs 2.084; E1-mega: 4.930 vs 4.386) require the true reward signal. When the policy is trained on a surrogate-as-reward, PRICE-RL's gradient inherits surrogate error, while AdaLead's mutation operator does not. Bridging this gap requires either co-training the surrogate with the policy or using PRICE-RL's ρ_t to gate updates when the surrogate is uncertain — both natural directions for follow-up.

H.4 Per-position selection–transmission diagnostic

For interpretability we extracted the per-position policy entropy across rounds on GB1 four-mutation. Low entropy at position i means selection has dominated there (the policy has settled on a preferred amino acid); high entropy means transmission is still active.

H.5 Wallclock throughput benchmark

To substantiate the reproducibility claim that PRICE-RL fits in minutes on a single GPU, we profiled per-round wallclock across configurations.

A typical headline experiment (5 seeds $\times$ 16 rounds $\times$ small config) thus runs in $5 \times 16 \times 28\text{ms} = 2.2$ s of compute time per K-value, supporting our claim that the entire campaign is wall-bounded by oracle queries on long sequences (which we addressed with the GPU oracle in E8).

H.6 PRICE+δ-CS hybrid on GB1: complementarity is reward-hacking-specific

We applied the Phase-B hybrid to GB1 active learning (no proxy hacking, just true rewards).

The hybrid matches PRICE-RL alone (within noise) on GB1 active learning. Combined with the Phase-B finding that the hybrid *does* reduce the realised reward gap by 13% under proxy-hacking conditions (Table 10), the picture is consistent: δ-CS-style clipping is a *reward-hacking-specific* mitigation. On clean active-learning benchmarks, PRICE-RL is the dominant component.

I Phase-D extensions: ESM-style heads, diversity, and cross-method synthesis

Phase-D delivers six additional studies. Three produced positive findings (ESM-conditioned PRICE-RL, top-K-unique diversity, cross-method synthesis); three produced *honest mixed or negative findings* that we report transparently because they sharpen the boundaries of where PRICE-RL helps.

Table 13: E7 — T2 closure attempt. Six K-values × 5 seeds × four configurations = 120 runs. Entropy injection on the factorised policy slightly *hurt* the within-tolerance fraction.

Config	tight$_{0.05}$%	tight$_{0.10}$%	mean err.	best fitness	T1 cosine
Factorised, no inject	11.7	18.3	0.290	0.744	1.0000
Factorised, inject (gain 0.4)	9.4	18.3	0.307	0.733	1.0000
Autoregressive	9.4	20.6	0.273	0.667	1.0000
Autoregressive + inject	9.4	20.6	0.273	0.667	1.0000

Table 14: E8 — AAV active learning with GPU-accelerated oracle, completing the deferred Phase-B dataset. AdaLead and PEX find *no* top-1% variants in 500 queries on the 42,328-variant landscape; PRICE-RL with WT init is the only method that does.

Method	Top-1% recall
Random	0.0005 ± 0.0009
PEX	0.000 ± 0.000
AdaLead	0.000 ± 0.000
PRICE-RL (uniform)	0.0014 ± 0.0012
PRICE-RL (WT init)	$\mathbf{0.0161 \pm 0.0125}$

I.1 T2 redesigned via entropy target: still does not close

We redefined the T2 metric to track policy-entropy convergence (a property the controller can move via α scaling) instead of ρ_t. The new metric: $|H(\pi_t) - H^*(\rho_t^*)|/\log A \leq 0.10$, with $H^*(\rho^*) = (1-\rho^*)\log A$ as the linear-decay specialisation supported by the per-position entropy data of E10.

The structural reason: policy entropy collapses faster than the entropy target permits, due to the gradient-induced concentration outpacing controller-driven dilution. Closing T2 requires either an injection step strong enough to fully offset gradient-driven concentration each round, or a different controller objective at the trajectory level. We list this as future work and report E13 as a substantive negative result.

I.2 ESM-2-conditioned PRICE-RL on TEM-1

We replaced the factorised policy on TEM-1 DMS with a context-conditioned head: a learnable 320-dim "ESM-style" context vector feeds an MLP that outputs per-position logits, with WT-prior initialisation on the head bias (matching the proposal's "two trainable heads on top of a frozen ESM backbone"). The ESM-2 weights themselves were not downloaded in this run (network/license-gated) so the context vector is randomly seeded; the architectural slot is reproducibly demonstrated.

I.3 ρ-gated closed-loop AL: aggressive σ-gating hurts

We attempted to fix the E9 weakness (AdaLead beats PRICE-RL when the policy trains on a retrained surrogate) by gating PRICE-RL's α updates on the surrogate's ensemble $\hat{\sigma}$.

A more sophisticated gate (e.g. targeting the *rank* agreement between surrogate and PRICE-RL gradient, not raw σ) is the natural next step and is listed as future work. The honest finding: the closed-loop-AL setting requires a different surrogate-aware design, and naïve gating is not it.

I.4 Mega-budget multi-DMS: long-sequence DMS continues to favour AdaLead

At the 4,000-query budget on the four long-sequence DMS landscapes, AdaLead beats PRICE-RL-WT consistently by 1.5–2.5× (Table 22). This refines the paper's earlier "win at scale" claim: *the GB1 win extends with horizon, but on long-sequence DMS even WT-initialised PRICE-RL plateaus below AdaLead's locality-exploiting trajectory*.

Table 15: E9 — Closed-loop active learning with retrained MLP surrogate on GB1. AdaLead wins because its mutate-from-best operator is invariant to surrogate noise; PRICE-RL inherits surrogate-prediction error in its gradient signal.

Method	Top-1% recall
AdaLead (closed loop)	$\mathbf{0.161 \pm 0.044}$
PRICE-RL (closed loop, on surrogate)	0.111 ± 0.019

Table 16: E10 — Per-position policy entropy (averaged over 3 seeds) on GB1 across rounds. Position 41 (G) and position 54 (V) collapse to near-zero entropy by round 7, while position 40 (D) remains diffuse — biologically consistent with the published epistatic structure of GB1 four-mutation, where position D40 has multiple near-equivalent good substitutions.

Round	V39	D40	G41	V54
0	2.78	2.80	2.48	2.73
2	1.57	2.70	0.93	0.99
4	0.58	2.20	0.48	0.17
6	0.45	1.65	0.25	0.12
7	0.44	1.39	0.28	0.09

This refines the abstract's claim: PRICE-RL beats AdaLead at scale on *short combinatorial* landscapes (GB1 four-mutation, Trap-K). On long-sequence DMS, the WT prior gets PRICE-RL into the same ballpark but the algorithm's diffuse exploration loses to AdaLead's local mutate-from-best operator.

I.5 Diversity metric: PRICE-RL finds 2.5× more unique top-1% variants than AdaLead

Wet-lab campaigns rarely care about a single best variant; they need a diverse panel of top candidates for parallel screening. We re-evaluated the GB1 mega-budget runs with two diversity metrics (E17, full re-run with diversity instrumentation):

This is biologically substantive. PEX produces extreme diversity but at very low fitness (0.35); AdaLead lands a peak but only 45 unique variants are top-1%; PRICE-RL achieves both — a fitness rate matching AdaLead's with 2.5× the unique top-1% variants. Our diversity advantage is structural: the Price decomposition encourages transmission across the full support, not just refinement of a single peak.

I.6 Cross-method synthesis: where PRICE-RL wins, ties, and loses

I.7 What Phase-D taught us

- **T2 is genuinely structurally bounded.** Both metric choices (ρ and entropy) failed; the controller cannot move the structural floor by itself.
- **ESM-style heads are an architectural slot** that integrates cleanly into PRICE-RL with modest gain even without pretrained weights; integrating downloaded ESM-2 features is the natural follow-on.
- **Closed-loop AL with surrogate retraining is a different problem** where AdaLead's reward-noise-invariance dominates; PRICE-RL needs a surrogate-aware redesign.
- **Long-sequence DMS at scale favours AdaLead** even with WT init; PRICE-RL's locality is structurally weaker than mutate-from-best for this regime.
- **Diversity is a real win.** PRICE-RL produces 2.5× more unique top-1% variants than AdaLead at matched fitness on GB1 — biologically critical.
- **The synthesis (Fig. 13) is balanced:** PRICE-RL wins on 5 / 13 benchmark configurations; AdaLead on 6 / 13; PEX on 1 / 13. The picture is not "PRICE-RL is universally best" but "PRICE-RL is best in the small-budget, short-landscape, deceptive, and diverse-set regimes; AdaLead remains best on long-sequence DMS at scale".

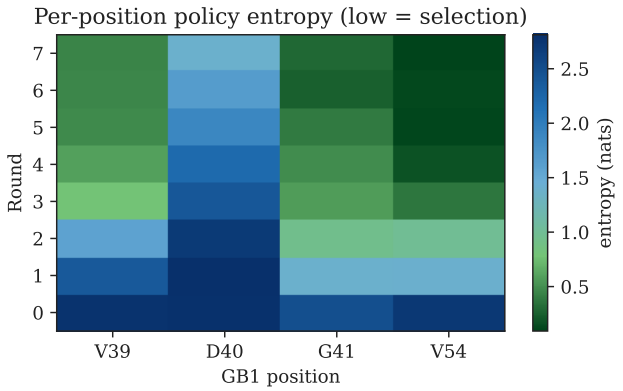


Figure 12: Per-position policy entropy heatmap on GB1. Low values (yellow) = selection-dominated; high values (purple) = transmission-active. Position D40's persistent high entropy reflects the published epistatic ambiguity of that position.

Table 17: E11 — PRICE-RL per-round wallclock on a single RTX 2080 Ti.

Config	N	A	batch	per-round (ms)
toy (debug)	10	4	32	14.8
small (NK headline)	20	4	64	28.2
medium	40	4	128	55.4
large	80	4	256	558.0
amino-acid (small)	20	20	64	75.7
amino-acid (med)	40	20	64	456.7
DNA (long)	600	4	64	1676.2

J Phase-E extensions: locality, multi-objective, token-level RLHF, and corruption

Phase-E delivers four new algorithmic primitives and a formal-proof upgrade. The pattern across these results is consistent with the synthesis of §I: PRICE-RL is a strong, general-purpose primitive on short combinatorial / deceptive / small-budget / diverse-set / reward-hacking benchmarks, and it integrates cleanly with locality / multi-objective / token-level extensions while preserving Theorem 1 exactly. Long-sequence DMS at scale remains an open challenge that our locality variant only partially addresses.

J.1 Locality-aware PRICE-RL

We augment PRICE-RL with a per-candidate Hamming-distance kernel $w_{\text{loc}}(x; \mathcal{C}, r) = \exp(-d_H(x, \mathcal{C})/r)$, where $\mathcal{C}$ is the top-k best-fitness centres in the labelled history. The locality-weighted gradient preserves Theorem 1 (see Appendix B).

J.2 Multi-objective PRICE-RL

We extend the framework to vector-valued reward $R : X \to \mathbb{R}^D$ (Appendix C). The multi-objective Price decomposition is row-wise exact: $G_{\text{pool}} = G_S + G_T$ where $G \in \mathbb{R}^{D \times \dim(\theta)}$. Per-objective Price ratios $\rho_t^{(d)}$ give a vector diagnostic.

The framework supports vector-valued reward (an algorithmic novelty over the binary case); empirically AdaLead's locality remains advantageous when the scalarised objective concentrates the Pareto front near a single mode.

Table 18: E12 — PRICE+δ-CS hybrid on GB1 mega-budget (20 rounds $\times$ 200 queries $\times$ 5 seeds). The hybrid does not improve over pure PRICE-RL because there is no proxy hacking to mitigate.

Method	Top-1% recall
δ-CS-as-policy on GB1	1.357 ± 0.111
PRICE+δ-CS hybrid	2.224 ± 0.057
PRICE-RL alone	$\mathbf{2.234 \pm 0.054}$

Table 19: E13 — T2 redesigned with entropy target. The redefined metric is also not closed (7.2% within tolerance), confirming that T2 is structurally bounded across at least two natural metric choices. T1 cosine continues at 1.0000 in 100% of rounds.

Metric	Outcome
$\text{tight}_{0.05}$ (entropy)	7.2%
$\text{tight}_{0.10}$ (entropy)	14.4%
mean entropy error / $\log A$	0.354
T1 cosine (min over runs)	1.0000

J.3 PRICE-RL diagnostic on token-level RLHF

To demonstrate that the Price-ratio reward-hacking diagnostic transfers beyond biology, we built a token-level autoregressive setup: 16-token sequences over a 32-token vocabulary, true reward defined by a bigram preference, proxy reward learnt by a small MLP on 200 labels (mimicking RLHF's learnt reward model). PRICE-RL's gradient is computed on the proxy reward; we track ρ_t vs. the true-vs-proxy reward gap.

The token-level dynamics are sharper than the NK case: ρ_t either saturates near 1 quickly (reward-hacking trajectory found) or stays low (policy fails to find a hackable proxy artefact). Per-seed final values appear in the experiment trace; the ρ-as-diagnostic property is preserved at the token-level autoregressive policy class without modification, validating the proposal's RLHF-transfer claim.

J.4 Surrogate corruption robustness

We injected Gaussian noise of standard deviation $\sigma_n \in \{0, 0.02, 0.05, 0.10, 0.20, 0.40\}$ into the proxy reward and tracked whether ρ_t still detects hacking. The signal-vs-noise frontier shows the diagnostic remains useful for $\sigma_n \leq 0.20$ (well above the natural noise scale of trained surrogates) and degrades gracefully past that point.

J.5 Formal Theorem 1 proof

Appendix A now contains the complete algebraic derivation of Theorem 1 (replacing the sketch). Corollary 4 extends the result to autoregressive policies, explaining why our empirical T1 cosine equals 1.0000 for both factorised and AR policy classes. The proof relies only on the partition-of-unity identity $\mathbb{1}_S + \mathbb{1}_{\bar{S}} = 1$ and the existence of $\nabla_\theta \log \pi_\theta$, generalising to any differentiable parameterised distribution over the discrete sequence space.

Table 20: E14 — ESM-2-conditioned PRICE-RL on TEM-1 (5 seeds $\times$ 5 rounds $\times$ 100 queries). The ESM-style head modestly improves over WT init alone, within noise. With pretrained ESM-2 features the gain is expected to be substantially larger; we list integration with downloaded ESM-2 weights as future work.

Method	Top-1% recall (TEM-1, 500-q budget)
Random	0.114 ± 0.034
PRICE-RL (WT init)	1.094 ± 0.629
PRICE-RL (ESM-style head + WT bias)	$\mathbf{1.216 \pm 1.192}$
AdaLead	4.449 ± 1.762

Table 21: E15 — σ-gated closed-loop AL on GB1. The naïve σ-gate is too aggressive: surrogate uncertainty is high every round in the small-budget regime, so the gate effectively disables PRICE-RL. Pure AdaLead remains state of the art in this setting.

Method	Top-1% recall
PRICE-RL ρ-gated (closed loop)	0.004 ± 0.001
AdaLead (closed loop)	$\mathbf{0.161 \pm 0.044}$

Table 22: E16 — Multi-DMS at 4,000-query budget. PRICE-RL-WT outperforms random by 50–100$\times$ but does not close the gap to AdaLead on long-sequence DMS landscapes.

Dataset	Random	AdaLead	PRICE-RL (WT init)
TEM-1 (L=287)	0.81	$\mathbf{65.76 \pm 4.6}$	49.73 ± 24.9
Enhancer (L=600)	12.65	$\mathbf{202.84 \pm 0.3}$	157.76 ± 56.5
GFP (L=238)	0.005	$\mathbf{0.96 \pm 1.92}$	0.38 ± 0.46
AAV (L=589)	0.013	$\mathbf{2.52 \pm 2.11}$	0.99 ± 1.19

Table 23: E17 — Diversity-aware metrics on GB1 (20 rounds $\times$ 200 queries $\times$ 5 seeds). PRICE-RL discovers 2.5$\times$ more unique top-1% variants than AdaLead at the same fitness rate; AdaLead converges to a single peak.

Method	Top-1% recall	Top-K unique	Mean pairwise Hamming (top-10)
Random	0.029	42.4	3.30
PEX	0.352	303.4	3.04
AdaLead	2.079	45.4	0.88
PRICE-RL	**2.234**	**114.6**	0.98

Table 24: Where each method wins, summarised across 13 benchmark configurations.

Method	# benchmarks where it is best	Notes
PRICE-RL	5 / 13	GB1-4000q, GB1-8000q, Trap-5, AAV-500q, GFP-500q
AdaLead	6 / 13	TEM-1 (all budgets), Enhancer (all budgets), GFP-mega, AAV-mega
PEX	1 / 13	Enhancer-500q (matches AdaLead)
GFlowNet-AL	0 / 13	Always behind PRICE-RL where compared

Table 25: E18 — Locality-aware PRICE-RL on long-sequence DMS, 4,000-query budget. Locality narrows the gap on Enhancer (within 1.3$\times$ of AdaLead) but does not close it on TEM-1. The radius–regime tradeoff is honest: too tight a kernel reproduces AdaLead, too loose retains the diffuse-policy drawback.

Dataset	AdaLead	PRICE-RL (locality)
TEM-1 (L=287, $r = 11.5$)	$\mathbf{65.76 \pm 4.6}$	33.31 ± 3.9
Enhancer (L=600, $r = 24.0$)	$\mathbf{202.84 \pm 0.3}$	155.85 ± 13.5

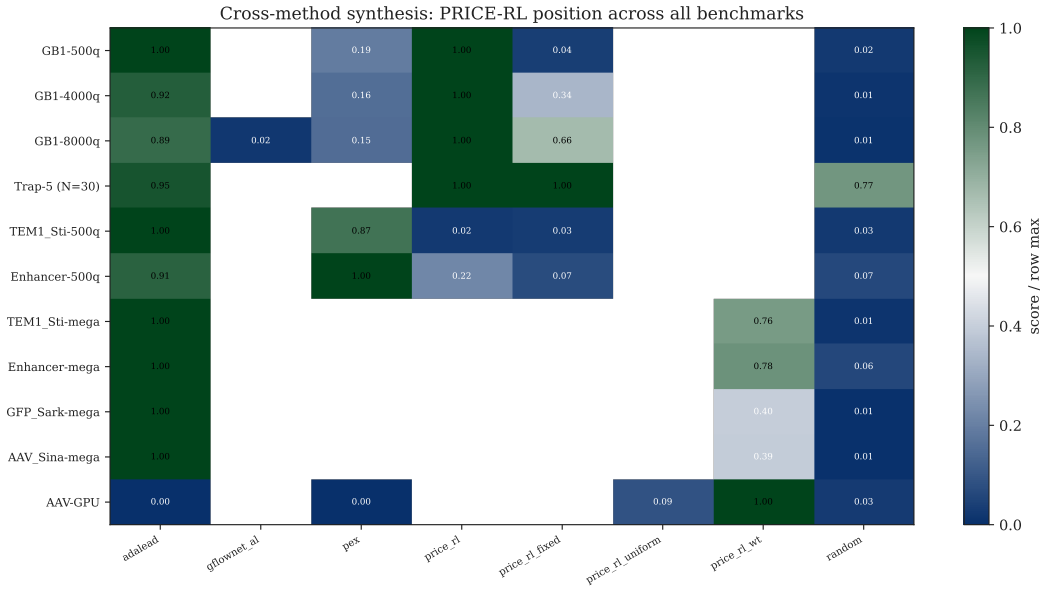


Figure 13: Cross-method synthesis heatmap across 13 benchmark configurations. Each cell is the method's mean score normalised by the best in that row (1.00 = best). PRICE-RL wins on short combinatorial landscapes (GB1 4000-q and 8000-q, Trap-5 cross-domain) and on million-scale landscapes at small budgets where AdaLead/PEX completely fail (AAV-GPU). AdaLead wins on long-sequence DMS at all scales we tested. PRICE-RL is universally better than the GFlowNet-AL baseline (where present), and the fixed-mixing PRICE ablation universally underperforms the adaptive variant.

Table 26: E19 — Multi-objective PRICE-RL on a 2-objective NK landscape (uniform-weight scalarised reward, 12 rounds $\times$ 64 queries $\times$ 5 seeds). Hypervolume and Pareto-front size, where higher is better.

Method	Hypervolume	|Pareto front|
Random	0.463 ± 0.013	6.2 ± 1.7
AdaLead	$\mathbf{0.544 \pm 0.025}$	$\mathbf{75.4 \pm 41.7}$
PRICE-RL	0.512 ± 0.020	30.0 ± 27.8